Master's Thesis

Multi Agent Reinforcement Learning-based Beam hopping for Interference Avoidance and Gateway Load Balancing in NTN

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The master thesis of Minsik Kim
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Abstract

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List of Abbreviations

LEO	Low Earth Orbit
UE	User Equipment
NTN	Non-Terrestrial Networks
TN	Terrestrial Networks
3GPP	3rd Generation Partnership Project
EMC	Earth-Moving Cell
CHO	Conditional Handover
BHO	Basic Handover
HO	Handover
MR	Measurement Report
SINR	Signal-to-Interference-plus-Noise Ratio
KPI	Key Performance Indicator
RSRP	Reference Signal Received Power
RLF	Radio Link Failure
UHO	Unnecessary Handover
HOPP	Ping-Pong Handover
ToS	Time of Stay

RB	Resource Block
RACH	Random Access Channel
OID	Orbital Information and Distance
FSPL	Free Space Path Loss
SIB	System Information Block
ISD	Inter-Site Distance
GPS	Global Positioning System
RRC	Radio Resource Control
ACK	Acknowledgment
TTT	Time-to-Trigger

Chapter 1. Introduction

1.1. Background and motivation

Low Earth Orbit (LEO) satellite communication has recently emerged as a promising next-generation technology for providing global coverage. The architecture of an LEO network consists of a multitude of satellites concurrently serving numerous terrestrial cells. However, this dense multi-satellite and multi-cell environment can lead to significant inter-cell interference. Furthermore, satellites require stable and high-capacity backhaul connections with gateways (GWs) to prevent data traffic bottlenecks and ensure seamless service for the cells they cover.

One of the most critical challenges in LEO satellite communications stems from the high mobility of the satellites. As satellites orbit the Earth at extremely high speeds, a ground terminal must frequently change its serving satellite. This high mobility necessitates frequent handovers and switchovers, which become a primary cause of degraded service continuity and can lead to temporary network disconnections.

In particular, conventional handover schemes based on received signal strength exhibit several limitations in this dynamic LEO environment.

Such schemes, which simply switch the connection to the satellite with the strongest signal, can trigger unnecessarily frequent handovers. Moreover, they often lead to the ping-pong effect, where handovers are rapidly repeated back and forth between two satellites at the cell edge. Consequently, the terminal's Time-of-Stay (ToS) with a single satellite becomes very short, which significantly degrades overall network efficiency and stability.

Therefore, to guarantee stable Quality of Service (QoS) and service continuity in LEO satellite networks, it is essential to develop intelligent handover and switchover schemes that can effectively manage the challenges posed by satellite mobility. This paper proposes a novel approach to address these pressing issues.

1.2. Contributions

To address the challenges of high mobility and dynamic network topology in LEO satellite networks, this paper proposes a novel handover and resource management scheme based on multi-agent reinforcement learning. The key contributions of this work are summarized as follows:

- **A Multi-Agent Reinforcement Learning (MARL) Framework for Joint Optimization:** We design a MARL framework where each satellite acts as an autonomous agent. These agents learn collaboratively to find the optimal policy for selecting both user cells and gateways simultaneously, adapting to the highly dynamic network environment without the need for a centralized con-

troller.

- **User Link Stability by Minimizing Handovers while Maximizing SINR:** Our proposed scheme enables the satellite agents to select user cells with a sophisticated dual objective. It aims to **maximize the Signal-to-Interference-plus-Noise Ratio (SINR)** for all served cells while concurrently **minimizing the total number of handovers**. This approach directly mitigates the ping-pong effect and enhances service stability for ground users.

- **Feeder Link Reliability by Preventing Bottlenecks and Reducing Switchovers:** The agents intelligently select gateways to **prevent traffic bottlenecks** in the backhaul network, ensuring efficient data routing. Crucially, this selection is performed while also **minimizing the frequency of feeder link switchovers**, which enhances the stability and reliability of the satellite-to-ground connection.

The effectiveness of our proposed scheme is validated through extensive simulations, which demonstrate significant improvements in network stability and user Quality of Service (QoS) compared to conventional, signal-strength-based methods.

1.3. Overview

The remainder of this thesis is structured as follows: Sec. II discusses handover, feeder link switching, and related research in beam hopping

and resource allocation. Sec. III presents the proposed multi-agent reinforcement learning framework in detail. Sec. IV discusses the simulation results and compares the proposed strategy with conventional approaches. Finally, Sec. V concludes the thesis.

Chapter 2. Related works

2.1. LEO satellite HO and FLSO schemes

In a NTN environment, the continuity of the service link between the UE and the satellite is ensured through HO. HO is the procedure by which a UE changes the serving cell; in terrestrial networks, this is typically triggered based on measurements of neighboring cell signals. However, in NTNs, the difference in received signal strength between satellites is often small, and thus distance- or time-based HO criteria are also employed. In addition to the service link, the continuity of the feeder link connecting the satellite to the ground GW is equally important. In particular, the presence or absence of a gNB onboard the satellite determines whether feeder link switchover entails a change in the service link. For a transparent satellite without an onboard gNB, a feeder link switchover changes the serving GW on the ground, which in turn changes the serving gNB, resulting in an HO. In contrast, for a regenerative satellite equipped with an onboard gNB, a feeder link switchover does not change the serving gNB, and therefore no HO occurs.

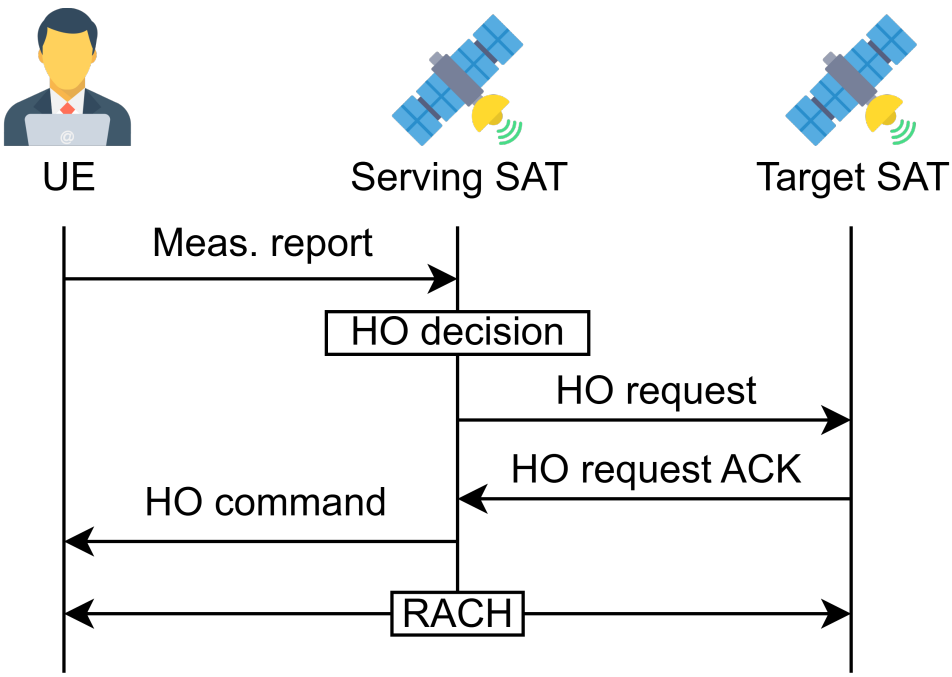


Fig. 2.1. Basic HO procedure

2.2. Reinforcement learning based Beam hopping and scheduling

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3.1. System model

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